

project-1

2025-09-26

Task 1

I am going to knit directly to a pdf file using the tinytex library.

A .Rmd file contains the R source code and text that is used to generate the .pdf file. The .pdf file is usually considered the final product.

```
medicaldata <- read.csv("medicaldata1.csv", header = TRUE)
head(medicaldata, n = 6)
```

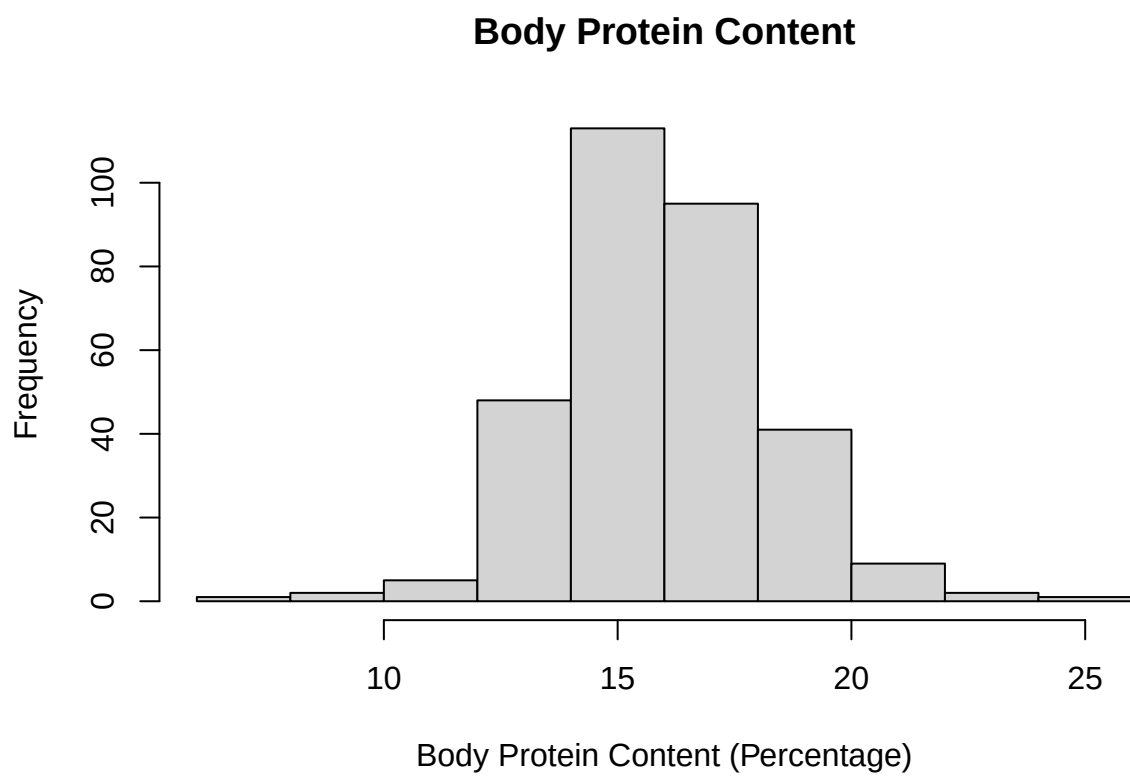
```
##   Age Height Weight Gallstone_Status Comorbidity CAD Diabetes_Mellitus
## 1  57   170  143.5             0           0  0             0
## 2  53   150  108.4             1           1  0             1
## 3  54   155   80.2             1           0  0             0
## 4  63   150   69.0             1           0  0             0
## 5  73   160  104.6             1           1  0             1
## 6  56   158  100.5             0           0  0             0
##   Total_Body_Water Total_Body_Fat_Ratio Lean_Mass Body_Protein_Content
## 1                66.2                42.30    57.70                7.99
## 2                40.0                50.92    48.99                9.05
## 3                32.8                46.13    53.87                9.63
## 4                29.9                42.46    57.39               10.49
## 5                52.7                34.20    65.77               11.31
## 6                35.9                49.80    50.25               11.41
##   Hepatic_Fat_Accumulation Glucose Triglyceride
## 1                        3      118          112
## 2                        2      119           97
## 3                        2      111          104
## 4                        2      106           96
## 5                        3      230          208
## 6                        3      129           81
```

Task 2

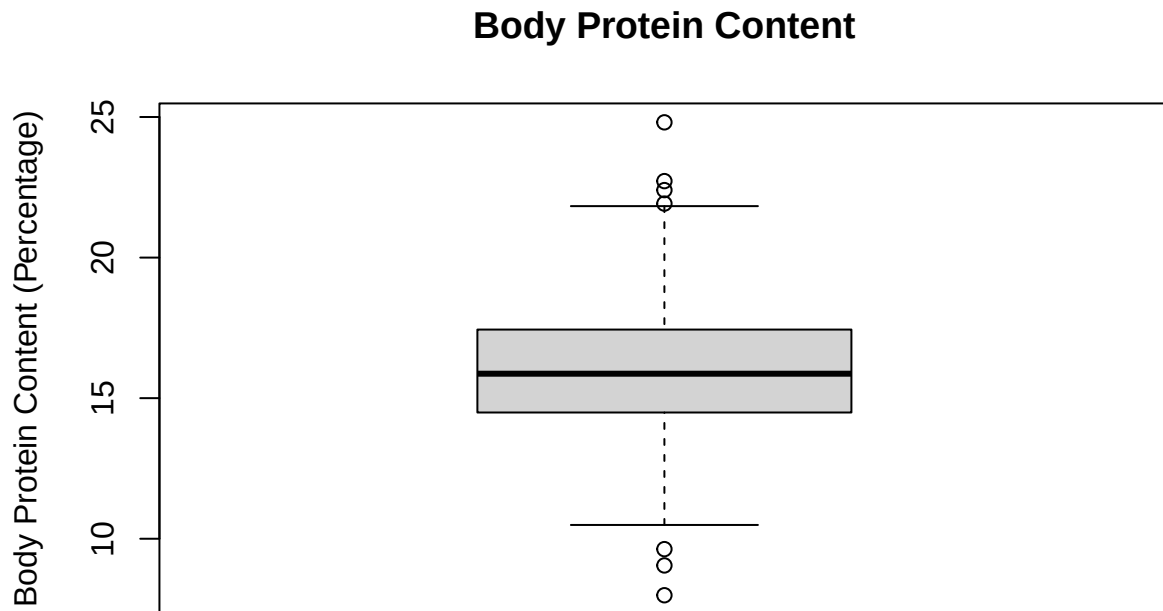
```
medicaldata$Gallstone_Status <- as.factor(medicaldata$Gallstone_Status)
medicaldata$Comorbidity <- factor(medicaldata$Comorbidity, ordered = TRUE, levels = c(0, 1, 2, 3), label = c("No", "Low", "Medium", "High"))
medicaldata$CAD <- as.factor(medicaldata$CAD)
medicaldata$Diabetes_Mellitus <- as.factor(medicaldata$Diabetes_Mellitus)
medicaldata$Hepatic_Fat_Accumulation <- factor(medicaldata$Hepatic_Fat_Accumulation, ordered = TRUE, levels = c(1, 2, 3), label = c("Low", "Medium", "High"))
```

Task 3

```
hist(medicaldata$Body_Protein_Content, main = "Body Protein Content", xlab = "Body Protein Content (Percentage)", ylab = "Frequency", col = "red", border = "black", las = 1)
```



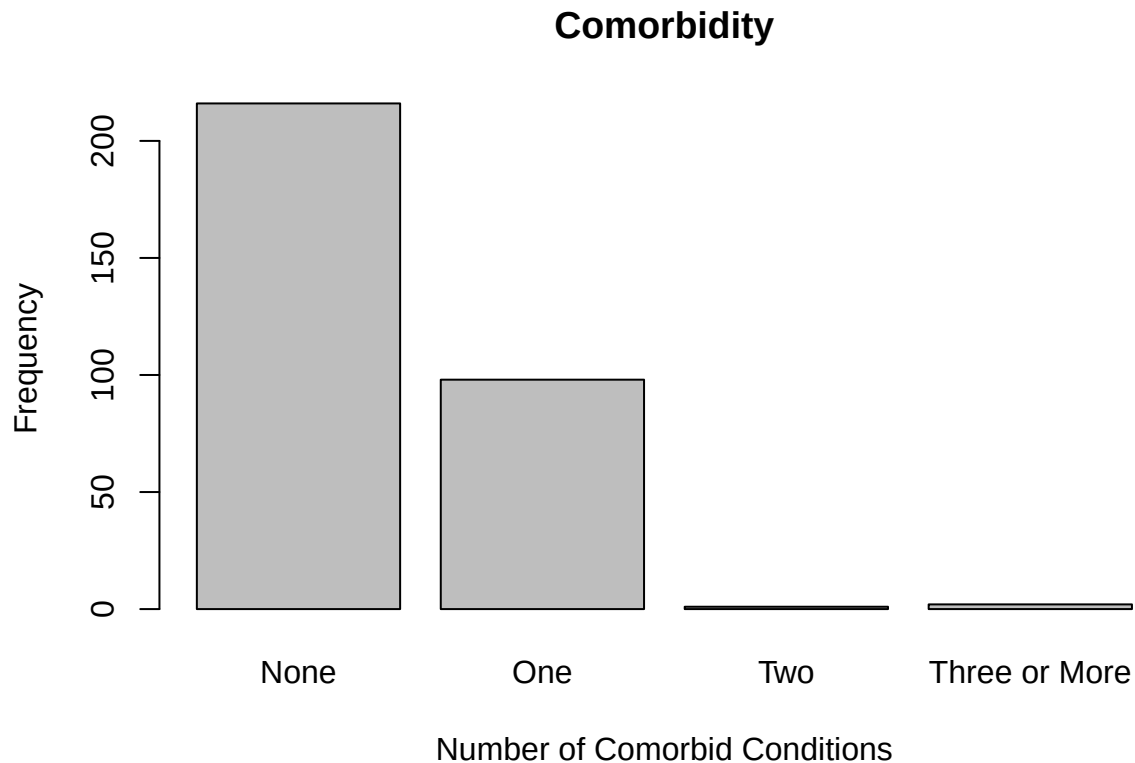
```
boxplot(medicaldata$Body_Protein_Content, main = "Body Protein Content", ylab = "Body Protein Content (Percentage)", col = "lightgray", border = "black", width = 0.8)
```



Based on the above boxplot and histogram, the body protein content appears to be normally distributed. The boxplot shows only 7 outliers out of a dataset of 317 individuals. The histogram is shaped like a bell curve, implying normality. The mean body protein content is 15.97095% with a standard deviation of 2.267927%.

Task 4

```
comorbidity_frequency <- table(medicaldata$Comorbidity)
names(comorbidity_frequency) <- c("None", "One", "Two", "Three or More")
barplot(comorbidity_frequency, main = "Comorbidity", xlab = "Number of Comorbid Conditions", ylab = "Frequency")
```



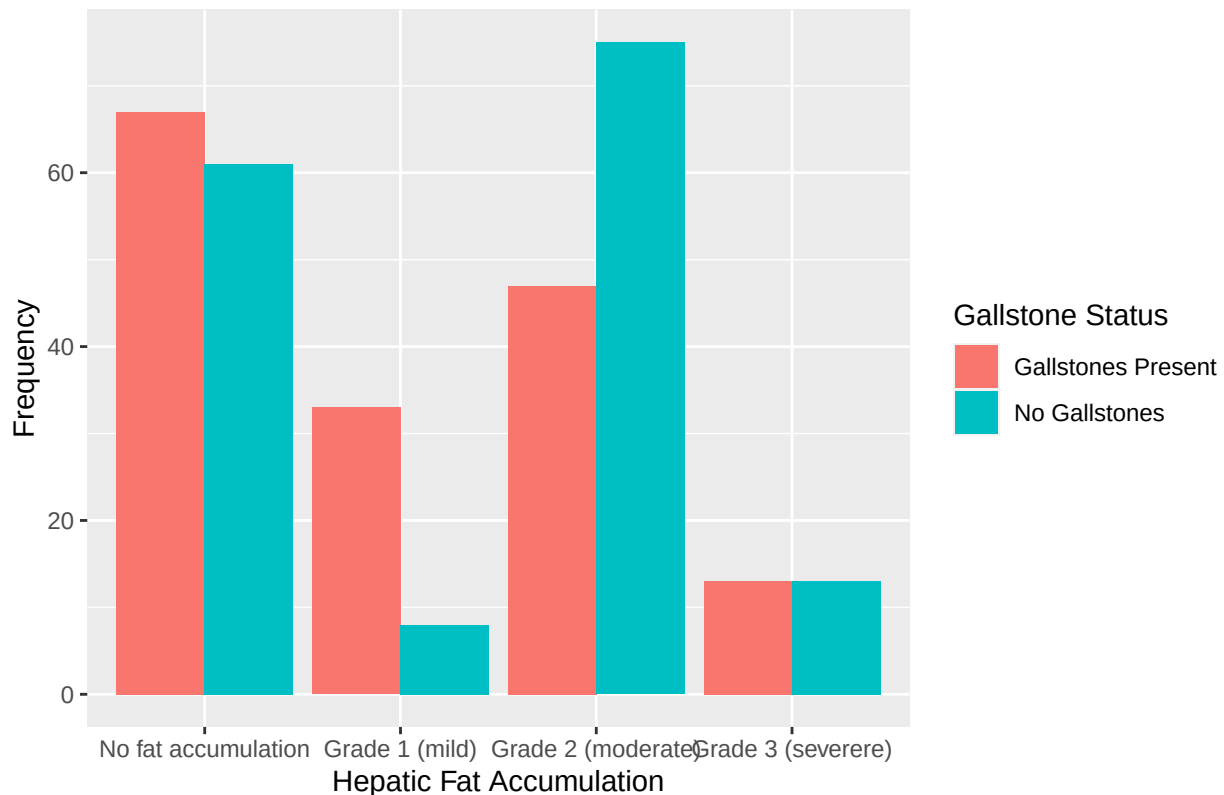
The graph of the number of comorbid conditions present in the individuals in the medical data set is skewed right. The most frequent number of comorbid conditions present is zero, which occurs 216 times. 98 individuals have one comorbid condition. 1 individual has two comorbid conditions. 2 individuals have three comorbid conditions.

Task 5

```
library(ggplot2)

ggplot(medicaldata, aes(x = Hepatic_Fat_Accumulation, fill = Gallstone_Status)) +
  geom_bar(position = "dodge") +
  labs(
    title = "Gallstone Status by Hepatic Fat Accumulation",
    x = "Hepatic Fat Accumulation",
    y = "Frequency",
    fill = "Gallstone Status"
  ) +
  scale_fill_discrete(labels = c("Gallstones Present", "No Gallstones"))
```

Gallstone Status by Hepatic Fat Accumulation



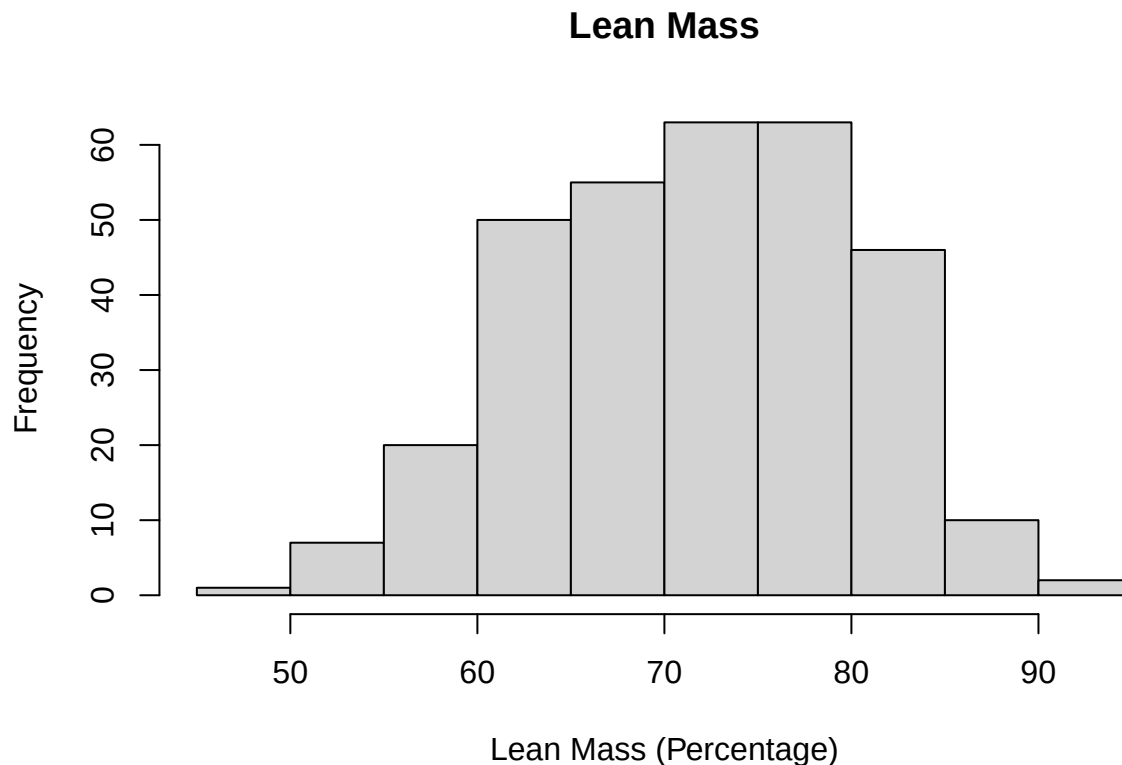
```
task5_distribution <- prop.table(table(medicaldata$Hepatic_Fat_Accumulation, medicaldata$Gallstone_Status))
task5_distribution
```

```
##
##              0      1
## No fat accumulation 0.21135647 0.19242902
## Grade 1 (mild)      0.10410095 0.02523659
## Grade 2 (moderate)  0.14826498 0.23659306
## Grade 3 (severe)   0.04100946 0.04100946
```

Based on the barplot shown above, it appears as if an individual's gallstone status does vary based on their hepatic fat accumulation. Based on the conditional probabilities, you have a 52.34375% chance to have gallstones given that you have no hepatic fat accumulation. This probability is 80.048781% given grade 1 hepatic fat accumulation, 38.52459% for grade 2 hepatic fat accumulation, and 50.0 % for grade 3 hepatic fat accumulation. These conditional probabilities vary greatly, leading to the conclusion that an individual's gallstone status does vary based on their hepatic fat accumulation.

Task 6

```
library(moments)
hist(medicaldata$Lean_Mass, main = "Lean Mass", xlab = "Lean Mass (Percentage)")
```



```
skewness(medicaldata$Lean_Mass)
```

```
## [1] -0.1179956
```

```
kurtosis(medicaldata$Lean_Mass)
```

```
## [1] 2.497133
```

```
shapiro.test(medicaldata$Lean_Mass)
```

```
##
```

```
## Shapiro-Wilk normality test
```

```
##
```

```
## data: medicaldata$Lean_Mass
```

```
## W = 0.99178, p-value = 0.07566
```

The skewness of the Lean Mass histogram is -0.1179956 and the kurtosis is 2.497133. These values as well as the fact that the histogram has a bell-curve shape implies normality. Running the Shapiro-Wilk normality test produces values of $W = 0.99178$ and a $p\text{-value} = 0.07566$. The null hypothesis is that the Lean Mass variable is normally distributed. The alternative hypothesis is that the Lean Mass variable is not normally distributed. Using a significance level of 3%, we can make the decision to not reject the null hypothesis because the $p\text{-value} = 0.07566 > 0.03$. We can conclude that there is enough evidence that the normality of the Lean Mass variable seems plausible.

Task 7

```
medicaldata$Age_Class <- cut(medicaldata$Age, breaks = c(-Inf, 40, 55, 65, Inf), labels = c("Younger",
```

```
boxplot(Total_Body_Fat_Ratio ~ Age_Class, data = medicaldata, main = "Total Body Fat Ratio by Age Class")
```



```
task7_summary <- aggregate(Total_Body_Fat_Ratio ~ Age_Class, data = medicaldata, FUN = function(x) c(mean(x), sd(x)))
```

```
task7_summary
```

```
##      Age_Class Total_Body_Fat_Ratio.mean Total_Body_Fat_Ratio.sd
## 1    Younger      25.183176      8.281911
## 2 Lower Middle      28.887266      7.593969
## 3 Upper Middle      30.818657      9.029098
## 4     Elder      28.802692      9.478935
##      Total_Body_Fat_Ratio.median Total_Body_Fat_Ratio.IQR
## 1              23.600000      12.800000
## 2              28.100000      11.300000
## 3              29.890000      13.750000
## 4              32.750000      14.385000
```

The mean body fat increases by age class until individuals reach Upper Middle age where the mean is at its highest of 30.818657%. It decreases slightly to a mean of 28.802692% as individuals reach Elder age. It should be noted that the Elder age class also has the highest standard deviation of 9.478935, which means there is a wider range of body fat ratios present in this age class. A general conclusion could be made that a person's total body fat ratio increases until they reach the Upper Middle age class, or between 55 and 65 years old, and then slightly decreases as they enter their elder years.

Task 8

```
t.test(medicaldata$Lean_Mass, mu = 70, alternative = "greater", conf.level = 0.94)

##
## One Sample t-test
##
## data: medicaldata$Lean_Mass
## t = 3.4041, df = 316, p-value = 0.0003747
## alternative hypothesis: true mean is greater than 70
## 94 percent confidence interval:
## 70.87668 Inf
## sample estimates:
## mean of x
## 71.61741
```

The null hypothesis is that the population mean is equal to 70. The alternative hypothesis is that the population mean is greater than 70. Since the p-value, 0.0003747, is less than the significance level, 0.06, we make the decision to reject the null hypothesis. We conclude that there is enough evidence present that the population mean is greater than 70. We are 94% confident that the true average mean exceeds 70.87668. The confidence interval is from 70.87668 to infinity, and the mean of Lean Mass is 71.61741, which falls inside the interval.

Task 9

```
glucose_cad_no <- medicaldata$Glucose[medicaldata$CAD == 0]
glucose_cad_yes <- medicaldata$Glucose[medicaldata$CAD == 1]

t.test(glucose_cad_no, glucose_cad_yes, alternative = "two.sided", conf.level = 0.99)

##
## Welch Two Sample t-test
##
## data: glucose_cad_no and glucose_cad_yes
## t = -1.4289, df = 10.315, p-value = 0.1826
## alternative hypothesis: true difference in means is not equal to 0
## 99 percent confidence interval:
## -92.46751 34.72758
## sample estimates:
## mean of x mean of y
## 107.6755 136.5455
```

The null hypothesis is that the population mean of Glucose is the equal for individuals with and without CAD. The alternative hypothesis is that the population mean of Glucose is not equal for individuals with and without CAD. Since the p-value, 0.1826, is greater than the significance level, 0.01, we make the decision to not reject the null hypothesis. We conclude that there is not enough evidence to suggest that the means differ. The 99% confidence level goes from -92.46751 to 34.72758. Since zero is in this range, it supports the conclusion that there is not enough evidence to suggest that the means differ.